

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Java** and **Kotlin** applications for point-of-sale terminals, delivering clean, maintainable code for embedded Android and AOSP-based POS devices.
- Integrated with **POS hardware** including **card readers**, **PIN pads**, and **NFC/contactless** modules by implementing and validating drivers and SDK integrations for PAX devices.
- Authored **payment flows** including authorisation, settlement, reversal, and offline-capable scenarios, ensuring correct handling of EMV, contactless, and magnetic stripe transaction types.
- Contributed to a **legacy transformation** programme by refactoring payment codebases to modern, modular architectures while preserving backward compatibility and minimising customer impact.
- Implemented secure coding practices to support **PCI PTS** and EMV kernel considerations, liaising with security teams on **key management interfaces**.
- Built and maintained automated **build, test, and release pipelines** for terminal images and application packages using **Jenkins**, supporting automated regression test runs on a device farm.
- Resolved critical production incidents by driving root cause analysis, producing fixes, and participating in post-release reviews, reducing mean time to resolution by approximately **30%**.

SKILLS

Android Development – Java, Kotlin, AOSP, POS Devices, PAX SDK, Embedded Android

Payment Systems – EMV, Contactless, NFC, Magnetic Stripe, Authorisation, Settlement, Reversal, Offline Processing

Security & Compliance – PCI PTS, EMV Kernel, Secure Coding Practices, Key Management, HSM

DevOps & CI/CD – Jenkins, Git, Gradle, Android Studio, Automated Testing, Device Farm, CI/CD Pipelines

Hardware Integration – Card Readers, PIN Pads, Peripherals, SDK Integration, Driver Implementation

General Software Engineering – REST APIs, Microservices, Agile Methodologies, Problem Solving, Debugging, Performance Optimisation

PROJECTS

POS Terminal Payment Application Modernisation

- Led the design and implementation of a new **Android POS application** using **Java** and **Kotlin**, focusing on modularity and maintainability for embedded devices.
- Integrated comprehensive support for **EMV**, **contactless**, and magnetic stripe transactions, ensuring seamless processing across various payment scenarios.
- Developed automated **unit and integration tests** using JUnit and Mockito, achieving over **85% code coverage** for core payment processing logic.

Automated Payment Certification Pipeline

- Engineered an automated testing framework in **Python** to simulate payment card scheme certification scenarios, reducing manual test execution time by **45%**.
- Developed scripts to parse and analyse logs from **PAX devices**, identifying and reporting on compliance deviations against **Visa** and **Mastercard** requirements.
- Integrated the pipeline with **Jenkins** for continuous integration, enabling rapid feedback loops for developers and QA teams during the certification process.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)